XRPL Validator — Daily Engineering Report · 2026-10-03

Period: 2026-10-03 00:00 to 24:00 US Eastern (2026-10-03 04:00 to 2026-10-04 04:00 UTC); ledgers #107,396,151 to #107,418,794. Every count from the ledger and the validator's logs stops at #107,418,794. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,622 of the 22,644 ledgers the network closed (99.903%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 22 during planned restarts.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 2 deployments (cycles 164, 165), 6 commits merged into main, 5 engine findings fixed (F412–F416).

Service levels

Measure	Value
Ledgers closed by the network	22,644
Validations issued	22,622
Coverage (validations / ledgers closed)	99.903%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	22
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	2
Basis: the hash the first peer announced as accepted (before cycle 183)	22,622

Restarts:

- 12:49:12 UTC: deployment of cycle 164; validations resumed at #107,404,480 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 18:52:10 UTC: deployment of cycle 165; validations resumed at #107,410,159 (10 ledgers unsigned between the last validation before the restart and the first after it)

Other planned maintenance:

- **107,410,183–#107,410,185 (3): Within cycle 165's start-up: the deployment completed at 18:52:58 UTC and the first health sample, at 18:56:14 UTC, still showed the validator loading its state.**

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,396,151–#107,418,794), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
164	`4729eb4`	12:50:00 UTC	1,701 passed / 0 failed; port 746 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	419 matched, 0 mismatched	#39 from 13:16 UTC: ended early: superseded by the next deployment (2026-10-03 18:52 UTC)
165	`83754c2`	18:52:58 UTC	1,961 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	418 matched, 0 mismatched	#40 from 19:19 UTC: CLEAN (2026-10-04 19:22 UTC)

Cycle 164 delivered: `4729eb4` fix(xrpl-ledger): F411 — the flow's dirFirst reads a relinked page link

Cycle 165 delivered: `eda5ffe` fix(xrpl-ledger): F412-F414 — AMM edges from testnet campaign 26; `4f74c0f` fix(xrpl-ledger): F415-F416 — ticket-paid TrustSet and EscrowCreate reserve; F414 zero-fraction tokens; `822fbd8` test(xrpl-node): campaign 26's three QualityIn/QualityOut/DeliverMin vectors are port-only; `31ea5ba` test(xrpl-ledger): amm_deposit_reserve_guard asks preclaim (finding 412 moved the gate); `634d8b6` test(xrpl-node): campaign 26 agent C — 58 multi-page book vectors (no finding); `83754c2` test(xrpl-node): 21 of agent C's multi-page book vectors are port-only

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
16:20 UTC	xrpl-validator	`eda5ffe`	fix(xrpl-ledger): F412-F414 — AMM edges from testnet campaign 26
16:20 UTC	xrpl-validator	`4f74c0f`	fix(xrpl-ledger): F415-F416 — ticket-paid TrustSet and EscrowCreate reserve; F414 zero-fraction tokens
16:20 UTC	xrpl-validator	`822fbd8`	test(xrpl-node): campaign 26's three QualityIn/QualityOut/DeliverMin vectors are port-only
16:20 UTC	xrpl-validator	`31ea5ba`	test(xrpl-ledger): amm_deposit_reserve_guard asks preclaim (finding 412 moved the gate)
16:45 UTC	xrpl-validator	`634d8b6`	test(xrpl-node): campaign 26 agent C — 58 multi-page book vectors (no finding)
16:45 UTC	xrpl-validator	`83754c2`	test(xrpl-node): 21 of agent C's multi-page book vectors are port-only

Findings fixed

- **fix(xrpl-ledger): F412-F414 — AMM edges from testnet campaign 26** (`eda5ffe`, xrpl-validator) F412: AMMDeposit's non-LP reserve gate is preclaim's (AMMDeposit.cpp:389-399), judged on the pre-fee balance; a depositor in (reserve, reserve + fee] deposits. F413: AMMClawback runs withdraw()'s holder reserve test before each send until fixCleanup3_4_0 (AMMWithdraw.cpp:690), measured by the issuer's preFeeBalance_. F414: a single-asset withdrawal whose tokens adjust to zero is refused (tecAMM_INVALID_TOKENS, …
- **fix(xrpl-ledger): F415-F416 — ticket-paid TrustSet and EscrowCreate reserve; F414 zero-fraction tokens** (`4f74c0f`, xrpl-validator) F415: TrustSet's reserve reads OwnerCount after consumeSeqProxy deleted the spending Ticket (TrustSet.cpp:330); we judge in preclaim, so the ticket comes off. F416: EscrowCreate's reserve, the same (EscrowCreate.cpp:431-445). F414: lpTokensIn whose fraction rounds to zero is zero tokens (refused), not 'no answer' (1e-15 USD single-asset withdrawal: tecAMM_INVALID_TOKENS).

Tests and campaigns

- **test(xrpl-node): campaign 26's three QualityIn/QualityOut/DeliverMin vectors are port-only** (`822fbd8`, xrpl-validator)
- **test(xrpl-ledger): amm_deposit_reserve_guard asks preclaim (finding 412 moved the gate)** (`31ea5ba`, xrpl-validator)
- **test(xrpl-node): campaign 26 agent C — 58 multi-page book vectors (no finding)** (`634d8b6`, xrpl-validator)
- **test(xrpl-node): 21 of agent C's multi-page book vectors are port-only** (`83754c2`, xrpl-validator)

Independence program

No independence record for UTC day 2026-10-03; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-03 15:35	10-03 16:20	xrpl-validator	`eda5ffe`	fix(xrpl-ledger): F412-F414 — AMM edges from testnet campaign 26
10-03 15:46	10-03 16:20	xrpl-validator	`4f74c0f`	fix(xrpl-ledger): F415-F416 — ticket-paid TrustSet and EscrowCreate reserve; F414 zero-fraction tokens
10-03 15:51	10-03 16:20	xrpl-validator	`822fbd8`	test(xrpl-node): campaign 26's three QualityIn/QualityOut/DeliverMin vectors are port-only
10-03 16:09	10-03 16:20	xrpl-validator	`31ea5ba`	test(xrpl-ledger): amm_deposit_reserve_guard asks preclaim (finding 412 moved the gate)
10-03 16:38	10-03 16:45	xrpl-validator	`634d8b6`	test(xrpl-node): campaign 26 agent C — 58 multi-page book vectors (no finding)
10-03 16:41	10-03 16:45	xrpl-validator	`83754c2`	test(xrpl-node): 21 of agent C's multi-page book vectors are port-only

Times in UTC. A commit appears on the day it was committed to main or merged into it.